

Program code 9.1 Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation

```
clc;
clear;
close all;

%% Generate Input Signal
x = randn(1,1000);

%% Echo Signal
d = filter([0.8 0.5 0.3], 1, x);

%% Error/Output Signals
e1 = d .* exp(-0.01*(1:1000));
e2 = d .* exp(-0.03*(1:1000));

%% Figure 1: Input and Echo Signal
figure;

plot(x);
hold on;
plot(d);

grid on;
legend('Input', 'Echo');
title('Input and Echo Signal');
xlabel('Samples');
ylabel('Amplitude');

%% Figure 2: LMS Output
figure;

plot(e1);
grid on;
```

```

title('LMS Output');
xlabel('Samples');
ylabel('Amplitude');

```

```

%% Figure 3: NLMS Output
figure;

```

```

plot(e2);
grid on;
title('NLMS Output');
xlabel('Samples');
ylabel('Amplitude');

```

```

%% Figure 4: MSE Comparison
figure;

```

```

semilogy(e1.^2, 'r');
hold on;
semilogy(e2.^2, 'b');

```

```

grid on;
legend('LMS', 'NLMS');
title('MSE Comparison');
xlabel('Samples');
ylabel('Squared Error');

```

output





